

YEAR 10 SCIENCE (EXTENSION)
TEST 1: MEASUREMENT
(Open Book)

NAME: SOLUTIONSMARK: 19

1. How many significant figures are involved in the following?

(a) 2130 kg 3(b) 0.00410 mm 3

(1 mark each)

(c) 120,000,000 km 2(d) 5.0 days 2

(4)

2. Convert the following to **standard units** and express your answers in **scientific notation**.(a) $5.53 \times 10^{-3} \mu\text{m}$ (b) 3.32×10^2 minutes $5.53 \times 10^{-9} \text{ m}$ $1.992 \times 10^4 \text{ s}$

(c) 0.0000235 MJ

(d) 4.35×10^2 tonnes $2.35 \times 10^1 \text{ J}$ $4.35 \times 10^5 \text{ kg}$

(1 mark each)

(4)

3. Perform the following calculations, expressing the answer to the correct number of significant figures.

(a) $31.4 + 0.049 - 8.91$ 22.5

$$\left[\begin{array}{l} \frac{1}{2} \text{ marks each} \\ \frac{1}{2} \text{ mark off each error.} \end{array} \right]$$
(b)
$$\frac{(3.23 \times 10^{-2})(4.1 \times 10^5)}{(9.689 \times 10^{-3})}$$
 1.4×10^6

(3)

4. The force on a current-carrying wire is given by the relationship:

$$F = kI$$

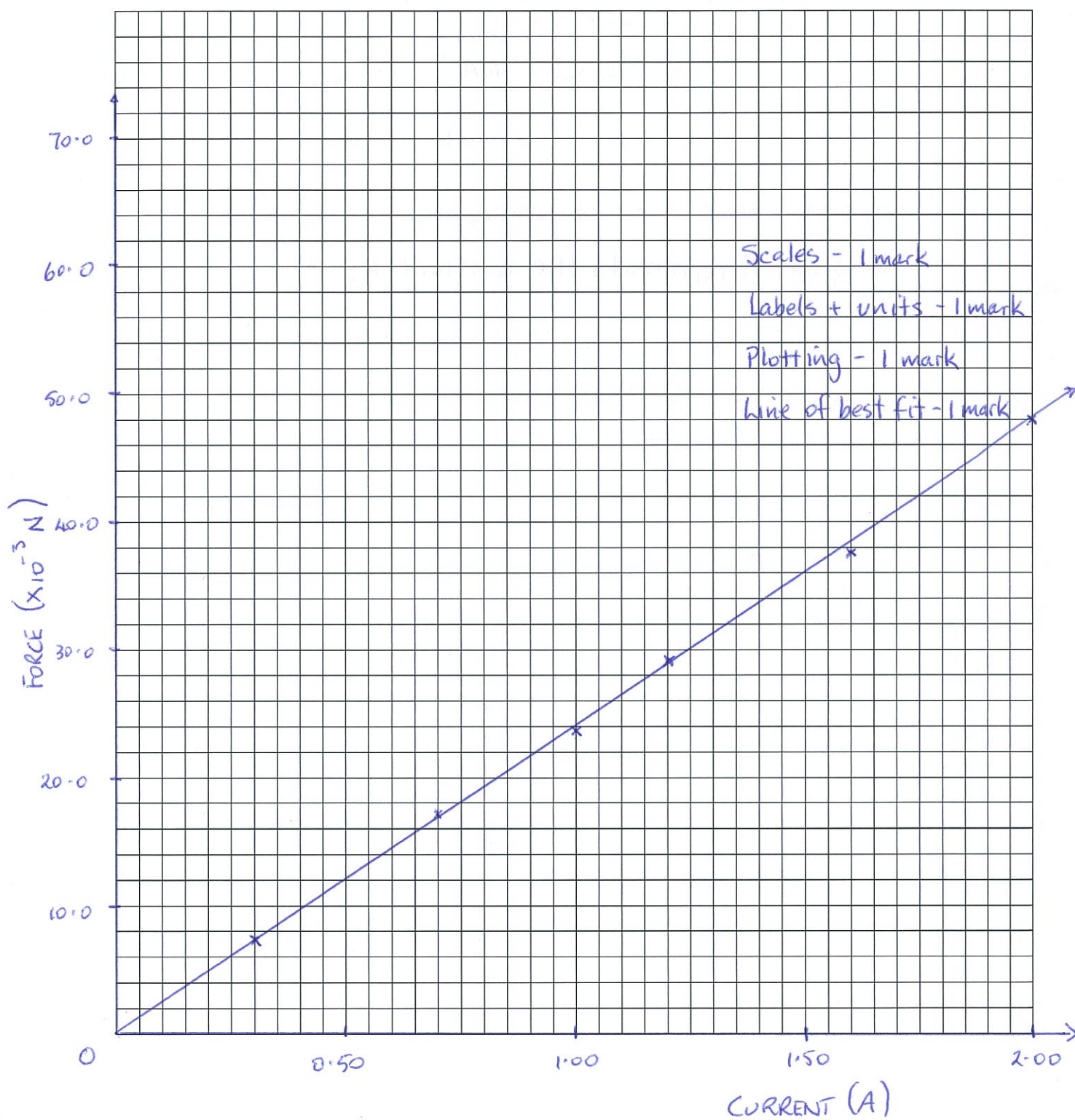
where F = the force experienced (N)
 I = the current flowing (A)
 k = a constant.

A group of students obtained the following data whilst investigating this relationship.

Current (A)	Force ($\times 10^{-3}$ N)
0.30	7.2
0.70	17.1
1.00	23.9
1.20	29.2
1.60	37.8
2.00	48.0

- (a) Graph the data on the grid provided and draw a line of best fit.

(4)



- (b) Determine the gradient of the line of best fit. Remember to include the appropriate units.

$$\text{gradient} = \frac{48.0 \times 10^{-3} - 0}{2.00 - 0} \quad (1)$$

$$= \frac{2.40 \times 10^{-2} \text{ NA}^{-1}}{\uparrow (1)} \quad (1)$$

(3)

- (c) What does the gradient represent in terms of the equation?

The gradient represents the constant k .

(1)